蘭州大學
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From Hardcore to Lightweight:

An AI-Assisted Multimodal Data-Curation Pipeline for Gorpcore Trend Intelligence

Assignment 2 — Project First Results

Milestone: Week 6 — Presentation 2

Group Number: 12

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Contents

1 Project Overview **2**

1.1 Introduction 2

1.2 Problem and Objective 2

1.3 Scope for This Milestone 2

1.4 What Makes Our Approach Different 3

2 Progress Summary **4**

3 Work Breakdown **4**

4 Challenges and Solutions **5**

5 Results and Findings **6**

5.1 Dataset v0 Obtained—Four Coordinated Sources 7

5.2 Text-Analysis Module 7

5.3 Consumer Pain Points Are Quantified, Not Guessed 8

5.4 Sentiment Is Overwhelmingly Neutral—a Useful Warning 8

6 Gorpcore-Curator-Agent Development **9**

6.1 Node A: Quality Gatekeeper Module 10

6.2 Node B: Vision Large Model Annotation Module 12

6.3 Node C: Semantic Cross-Checking Module (In Progress) 14

6.4 Node D: Human Review Module (In Progress) 15

7 Next Steps **15**

7.1 Feasibility and Risk 15

7.2 Engagement and Responsiveness 16

8 Conclusion **17**

Appendix **18**

Realised Micro-Ontology 18

1. Project Overview

1.1 Introduction

This document is the Week 6 progress report for Group 12's INFO 202 project—a checkpoint between the Week 3 problem-and-plan presentation and the Week 9 final delivery. It follows the project-status structure requested in the Assignment 1 feedback: overview, progress summary, work breakdown, what has changed, early results, and next steps. The project sits squarely within the course theme of trend intelligence for brands through AI-assisted data curation. Its application domain is the Chinese sportswear and outdoor-fashion market, specifically the transition of the Gorpcore aesthetic from hardcore mountain gear toward a lighter, city-ready style [2, 3], and the question it answers is the one posed in the project brief: how can a brand use public, multimodal data to detect fashion trends in a defined period and turn them into concrete recommendations on colours, materials, silhouettes and functional details [5]?

1.2 Problem and Objective

Hardcore outdoor apparel is engineered for extremes, but in the daily commute of China's first- and second-tier cities those same garments are often experienced as hot, heavy and visually "too much." Brands such as Anta and Li-Ning are pushing into the urban-outdoor segment but lack a reusable, evidence-based way to learn which design elements consumers actually reward and which they reject [3, 4]. The objective of this project is to build an AI-assisted curation pipeline that converts fragmented, noisy, multimodal public data into a structured trend-intelligence asset and a set of differentiated design recommendations for a "lightweight Gorpcore" product line.

Figure 1. Visual reference for the lightweight Gorpcore aesthetic.

1.3 Scope for This Milestone

For Week 6 our scope is the curation backbone rather than the final recommendations. Concretely, this milestone covers: (i) a multi-source raw dataset (Dataset v0); (ii) a metadata schema and version control that make the data reusable; (iii) significant progress, successful implementation and future scheme design of the curator agent; and (iv) early descriptive findings on pain points, sentiment and visual attributes. The semantic cross-check at scale, the visual word cloud and the final design white paper are scheduled for Weeks 6–9 and described in Section 7.

1.4 What Makes Our Approach Different

A recurring risk in this kind of project is to stop at "what is popular," producing a dashboard of trending colours that any scraper could generate. We deliberately differentiate on three points:

- **Pain-point-to-design bridging.** We do not only record what people like; we mine negative and critical signals (returns, complaints, "too hot," "too heavy") and map each pain point to a concrete design implication and candidate elements. This turns curation into an actionable brief, not a popularity chart.
- **Multimodal cross-verification.** Visual attributes extracted from images are cross-checked against the text that accompanies them. When an image of a heavy fleece shell is described as "perfect for summer commuting," the record's confidence is lowered. This filters the "filtered, idealised" bias of lifestyle platforms.
- **A controlled domain vocabulary.** Colours, materials, cuts, functional components, scenarios and pain points are normalised against a project micro-ontology, so that signals from four very different platforms become comparable and reusable.

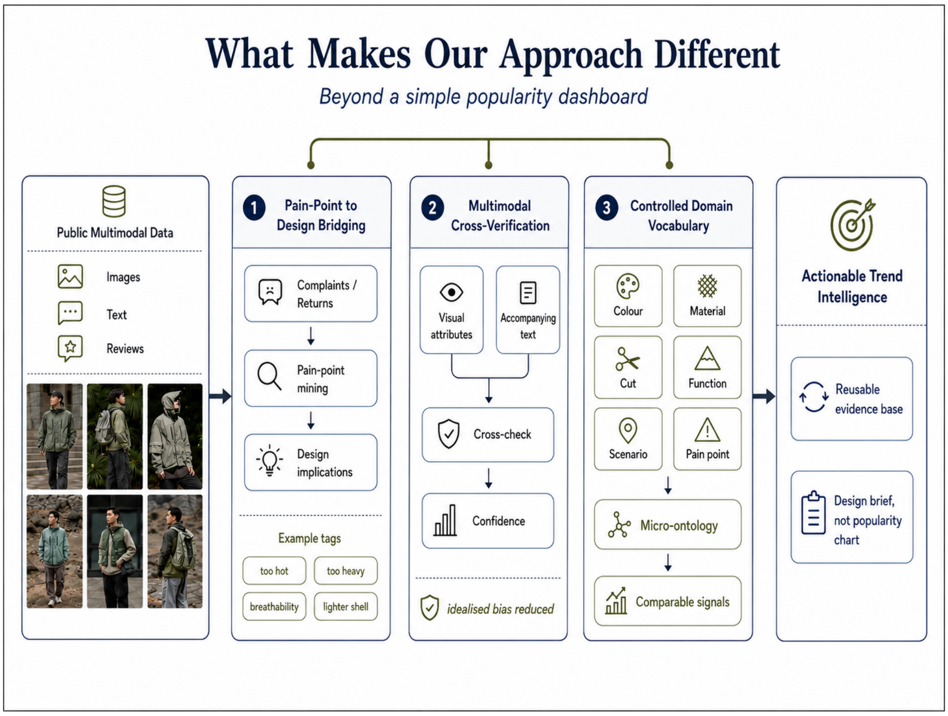

Figure 2. Project overview.

2. Progress Summary

Against the four-phase roadmap proposed in Assignment 1, the project is on schedule. Phase 1 (acquisition and pre-processing) is essentially complete; Phase 2 (multimodal feature engineering) is partially complete, with the text track finished on the full corpus and the image track validated on pilot batches; Phase 3 (automated curation and visualisation) is in progress; Phase 4 (strategy and delivery) is planned. Table 1 summarises module-level status.

Table 1. Module-level progress summary.

Module	Planned Deliverable	Status	Evidence
1. Data acquisition	Dataset v0 + metadata list	Complete	4 scrapers; 39,611 text records; 1,077 images
2. Cleaning & desensitisation	Dataset v1 (compliant)	In progress	Text cleaning done; face-blur pending
3. Multimodal features	Structured feature DB	Partial	Text track complete; vision pilot annotated
4. AI automated curation	Curation report + visual map	In progress	Agent base completed
5. Design strategy	Lightweight Gorpcore plan	Planned	Pain-point→design mapping drafted
6. Report delivery	Final report + materials	Planned	This progress report

3. Work Breakdown

Data Acquisition. Builds and runs the four-platform collection layer (Xiaohongshu, Bilibili, Taobao and JD) and consolidates the raw dataset, ensuring every record carries consistent provenance information before hand-off.

Data Governance. Turns the raw data into a governed, reusable dataset through image desensitisation, removal of personally identifiable information, metadata standardisation across platforms, and construction of a unified master table.

Curation Agent. Runs the image-side multi-node agent: a quality gatekeeper that filters out weak and duplicate images, a vision annotator that produces structured visual labels, a semantic cross-check against the text features, and a human-in-the-loop step that reviews low-confidence records before they enter the curated dataset.

Text-Analysis Module. A standalone module that processes the full text corpus: cleaning, tokenisation and word-frequency analysis, pain-point extraction, sentiment scoring, construction of a domain micro-ontology, and mapping of pain points to design implications.

Cross-Modal Fusion and Visualisation. Extracts colour and garment-component features, generates image embeddings, and fuses the image labels, text features and governed dataset into a structured feature database that drives the early-insight visualisations.

4. Challenges and Solutions

Several practical factors compelled us to adjust the plan for Assignment 1. We present these circumstances candidly, outlining our current remedial measures as well as potential further improvements to be implemented subsequently.

1. **Challenge 1: JD Collection Blocked.**
   Our JD review collector returned no usable records this milestone due to anti-automation measures on the platform.
   - *Current solution — Source Rebalanced:* Rather than over-invest in defeating those measures, we rebalanced the e-commerce signal onto Taobao (481 competitor reviews) and leaned more heavily on Bilibili comments and danmaku, which proved to be a richer, blunter source of consumer criticism. JD remains a stretch goal but is no longer on the critical path.
   - *Future improvement:* In the future, we may choose to use third-party APIs to scrape relevant data from JD.
2. **Challenge 2: Anti-Scraping and Session Handling.**
   Platforms aggressively detect automation.
   - *Current solution — Stealth + Cached Auth:* We adopted a stealth browser configuration and a cached authentication state so that sessions remain stable, and we throttled request rates to stay within respectful limits.
   - *Future improvement:* Third-party aggregation API.
3. **Challenge 3: Review Bias on E-Commerce Platforms.**
   Merchants routinely inflate ratings through post-purchase incentives, and platforms automatically assign five-star defaults to buyers who do not respond within a set period. As a result, many records retrieved as low-star reviews are platform-generated default entries—short, contentless placeholders—that carry a negative label but contain no usable design signal.
   - *Current solution:* The text-analysis module applies a minimum character-count threshold during text cleaning to exclude default-template reviews from pain-point extraction and sentiment analysis.
   - *Future improvement:* Incorporate a credibility score combining review length, posting timestamp, and verified-purchase status to further reduce platform-induced noise. Increase data scraping volume.

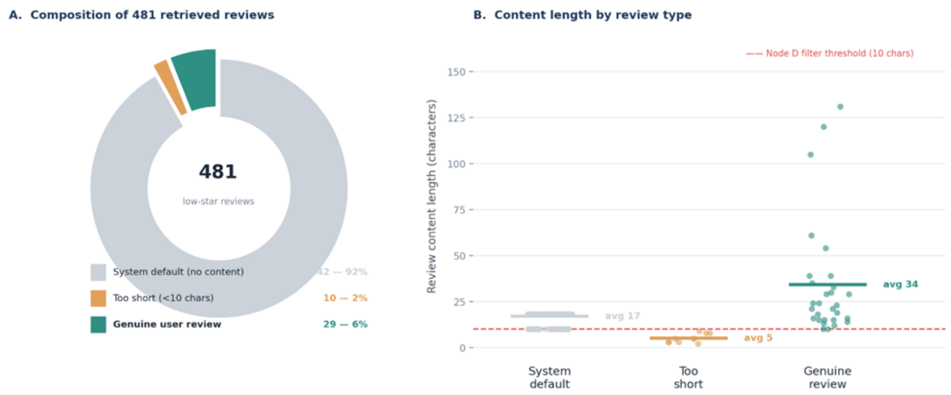

Figure 3. Review bias on e-Commerce platforms.

4. **Challenge 4: VLM Cost, Latency, and Complex Data Source Processing.**
   Full-scale annotation of more than 1,000 images is slow and costly. The larger challenge is that raw social-media images are noisy: they include landscapes, product advertisements, collages, half-body shots and full-body outfit images.

   - *Current solution — Gate-First Filtering and Confidence-Gated Batching with Dual-Stage Prompting:* We implemented a pipeline strategy that combines hierarchical curation with gated batching. Initially, a lightweight local gating algorithm is deployed, efficiently intercepting near-duplicate images and obvious non-outfit content based on low-level visual features. Subsequently, we designed a rigorous and comprehensive dual-stage prompt engineering framework for the VLM. The complete solution is described in Section 6.

5. **Challenge 5: Noisy Text Statistics.**
   Our first word-frequency pass was dominated by stop-words, numerals and laughter tokens, which carry no design signal.

   - *Current solution — Stronger Stop-Word and Domain Weighting:* The next iteration of the text-analysis module adds a stronger stop-word list, removes platform-specific filler tokens, and weights terms toward the domain micro-ontology so that material, colour, scenario and pain-point terms surface more clearly. At the same time, potentially meaningful social-language signals such as hashtags, slang and complaint phrases are preserved rather than removed automatically.

5. Results and Findings

The findings below are early and descriptive, but they are produced by the live pipeline on real data rather than asserted from the literature.

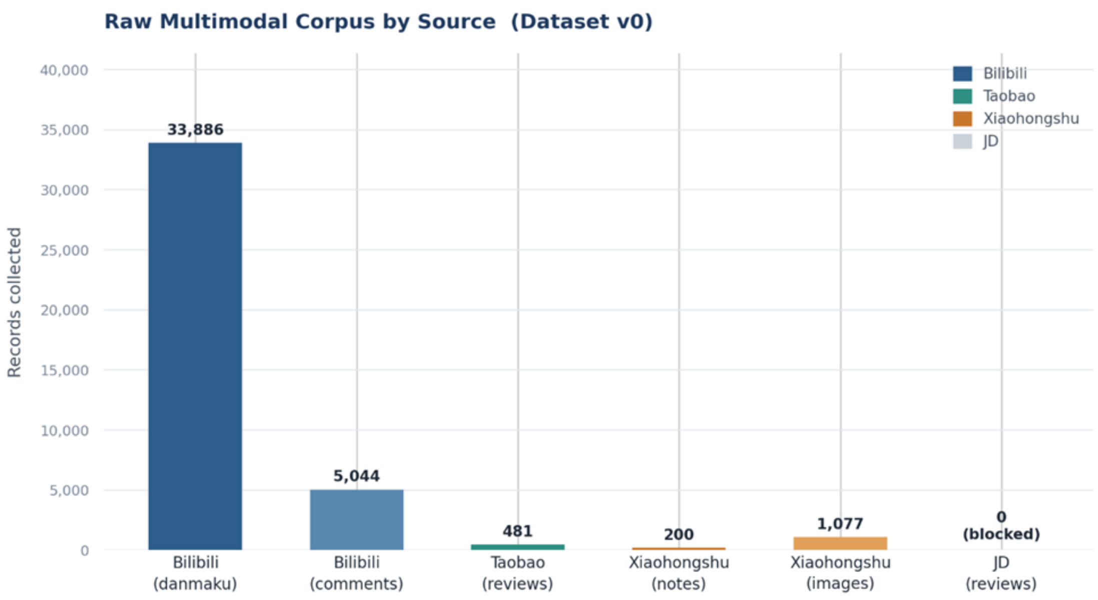

Figure 4. Raw multimodal corpus collected per source (Dataset v0).

5.1 Dataset v0 Obtained—Four Coordinated Sources

We implemented four independent collectors, each targeting a different curation value. Xiaohongshu supplies the aspirational visual side (outfit images and captions); Bilibili supplies dynamic usage feedback and blunt criticism through video comments and bullet-chat ("danmaku"); Taobao supplies competitor product reviews; JD was planned as a second e-commerce source. The collectors for Xiaohongshu and Bilibili use a Playwright browser-automation stack with a stealth layer and a cached login state, while the e-commerce reviews are pulled through a third-party data SDK [8]. Figure 4 shows the resulting raw volume by source.

Beyond volume, each source is captured in a structured, traceable format with consistent provenance fields, so that records from four very different platforms remain comparable downstream. Table 2 compares the four sources, including their collection methods.

5.2 Text-Analysis Module

The text-analysis module is a standalone component. Its role is to process the multi-platform text corpus, including Xiaohongshu captions, Bilibili comments and danmaku, and Taobao competitor reviews.

The pipeline includes text cleaning, duplicate removal, template-review filtering, jieba tokenisation, stop-word filtering, word-frequency statistics, pain-point keyword extraction, lexicon-based sentiment scoring, construction of a six-facet micro-ontology, and pain-point-to-design-implication mapping [5]. During cleaning, meaningless filler tokens, numerals and repeated laughter expressions are removed, while potentially meaningful social signals such as hashtags,

Table 2. Data-source comparison: collection method and raw volume.

	Xiaohongshu	**Bilibili**	**Taobao**	**JD**
Role	Primary visual source	Candid criticism	E-commerce evidence	Planned 2nd e-commerce
Method	Playwright + stealth + cached login	Playwright + stealth	Third-party SDK	Third-party SDK
Query	#Gorpcore, #functional-style (100 each)	"Gorpcore," "functional-style"	6 competitor queries (Arc'teryx, Salomon, Patagonia)	Same as Taobao
Format	JSON + .webp images + image_index.csv	Single structured JSON	JSON (summary + reviews)	JSON
Key fields	note_id, raw_text, image_urls, downloaded_image_paths, publish_time	summary, pain_point_keywords, all_danmaku, all_comments	Review text, star rating, brand	—
Raw volume	200 notes / 1,077 images	33,886 danmaku / 5,044 comments	481 reviews (1–2 star only)	0 (blocked)

slang, scenario words and complaint phrases are preserved.

The module outputs cleaned text, tokens, word frequency, a pain-point table, sentiment labels, ontology tags, pain-point-to-design mapping and text feature vectors. These outputs feed the downstream fusion and visualisation layer. Because the current method is mainly keyword- and lexicon-based, its results are treated as early descriptive signals rather than final ground truth. Sarcasm, slang and platform-specific expressions may still be under-detected, so the next stage will include manual validation and ontology refinement.

5.3 Consumer Pain Points Are Quantified, Not Guessed

The text-analysis module extracted and de-duplicated pain-point mentions using a predefined keyword set aligned with the project micro-ontology. The current results should be interpreted as keyword-based complaint signals rather than a complete measurement of consumer dissatisfaction. Within this cleaned corpus, price appears as the most frequent complaint signal, followed by breathability. Commute suitability, durability, weight, fit and zipper reliability form a longer tail. These signals provide an empirical starting point for the "lightweight, city-ready" design direction, but they will be validated against image labels and manual samples in the next stage.

5.4 Sentiment Is Overwhelmingly Neutral—a Useful Warning

Lexicon-based scoring labelled 6.8% of the cleaned text records as positive, 92.3% as neutral and 0.9% as negative. The difference between the raw text count and the sentiment-labelled count comes from records removed during cleaning, such as empty, duplicated or template-like entries.

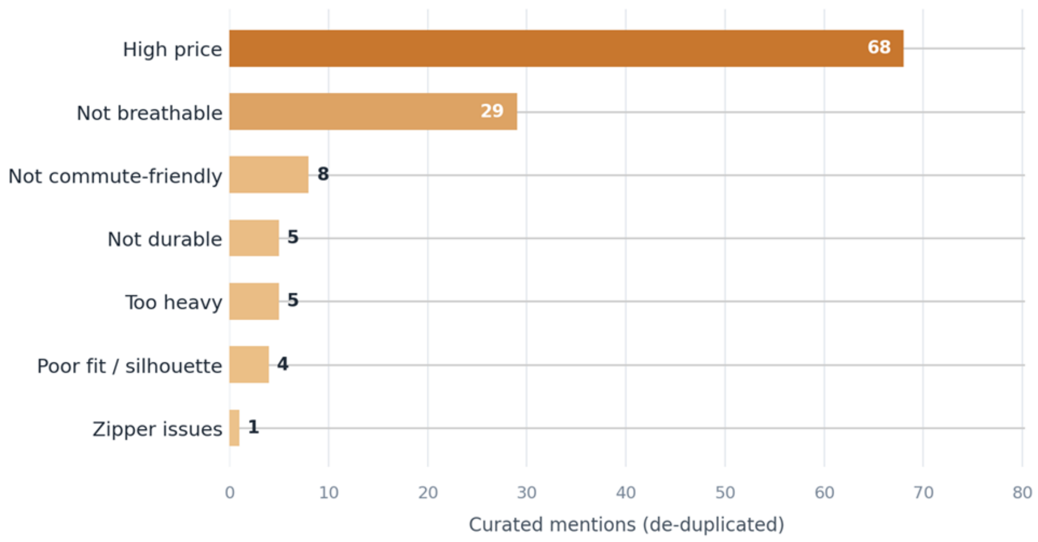

Figure 5. Curated, de-duplicated consumer pain-point signals.

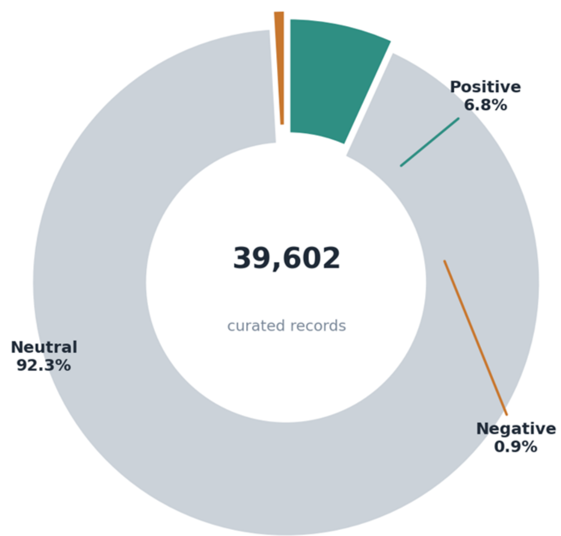

Figure 6. Sentiment distribution over 39,602 curated text records.

The large neutral share should not be read as "consumers have no attitude." Instead, it suggests that much of the collected social text is descriptive, short or context-dependent, which makes simple sentiment scoring conservative. Because sarcasm, slang and platform-specific expressions can be missed by lexicon-based methods, sentiment is used here as a weak prioritisation signal rather than a final judgement. The negative subset remains useful because it highlights records that deserve closer pain-point analysis and manual checking.

6. Gorpcore-Curator-Agent Development

We carried out the core development task of the `Gorpcore-Curator-Agent`. The objective is to build an AI-assisted data curation pipeline that performs quality screening, visual annotation, semantic consistency checking, and human review on multimodal Gorpcore outfit data collected from platforms such as Xiaohongshu. This provides a structured data foundation for subsequent trend analysis, design element extraction, and brand design recommendations.

The agent is designed to do more than send images to a VLM. It records why each image is accepted or rejected, stores per-image JSON labels, logs annotation errors, and prepares low-confidence cases for later human review. The overall architecture is shown in Figure 7.

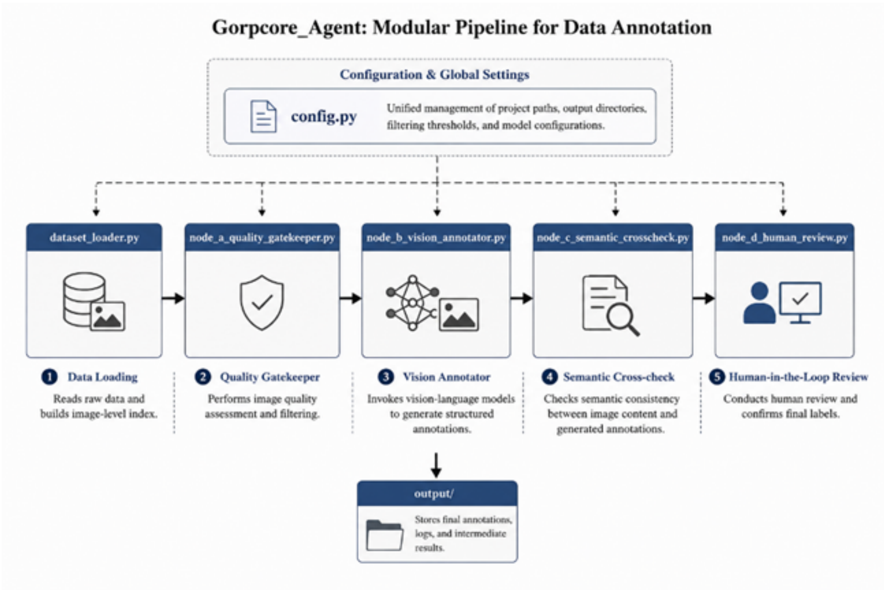

Figure 7. Overall architecture of the `Gorpcore-Curator-Agent`.

This modular design gives the agent strong maintainability. Each node has clearly defined inputs and outputs, allowing it to be tested independently or connected into a complete pipeline. The agent development progress, output results and related logs are fully available at: `https://github.com/De-Carl/Gorpcore_Project/tree/main/Gorpcore_Agent`.

6.1 Node A: Quality Gatekeeper Module

Node A selects images from Dataset v0 that are suitable for visual annotation. Since Xiaohongshu images are highly diverse and may include pure landscapes, duplicates, or irrelevant content, the system does not send all images directly to the multimodal model. Instead, it uses a hierarchical screening process to reduce API costs and prevent invalid images from affecting later Gorpcore outfit analysis.

First, Node A expands the original Xiaohongshu note data into image-level records. Since one note can contain multiple images while the subsequent analytical unit is a single image, the programme generates a unique `Image_ID` for each image and preserves contextual information such as the source note, keyword, post text, and image path.

Figure 8. Node A quality-gatekeeper processing pipeline.

Node A then performs a basic quality check. It uses an image-processing library to forcibly read the underlying image data stream in order to intercept corrupted files, while also setting aspect-ratio and minimum-pixel thresholds to remove thumbnails and abnormally long screenshots.

For duplicate-image filtering, Node A uses the average-hash perceptual hashing algorithm. Each image is resized to 64×64 pixels, converted to greyscale, and transformed into an image fingerprint based on average brightness. The system then compares the Hamming distance between the current image and previously accepted images. If the distance is below the preset threshold, the image is treated as a near duplicate and isolated. This mechanism helps reduce duplicate samples and feature redundancy in later statistical analysis.

Node A further optimises the filtering of landscape-like images. Since real Gorpcore outdoor outfit photos often contain large natural backgrounds, they can be confused with pure landscape images. To address this, the system combines HSV colour analysis with YOLO human detection. HSV analysis is used to identify green vegetation and blue sky or water regions and calculate the natural-colour ratio, while YOLOv8n detects whether a person is present and estimates the human area ratio. Node A blocks an image with high confidence only when no person is detected and the natural background ratio is extremely high, or when the person is very small and natural elements dominate. Borderline cases are passed to Node B for further semantic judgement, reducing the risk of removing valuable outfit samples. Thanks to the algorithm we designed, the agent was able to filter out some unnecessary landscape images, as illustrated in the Figure 9.

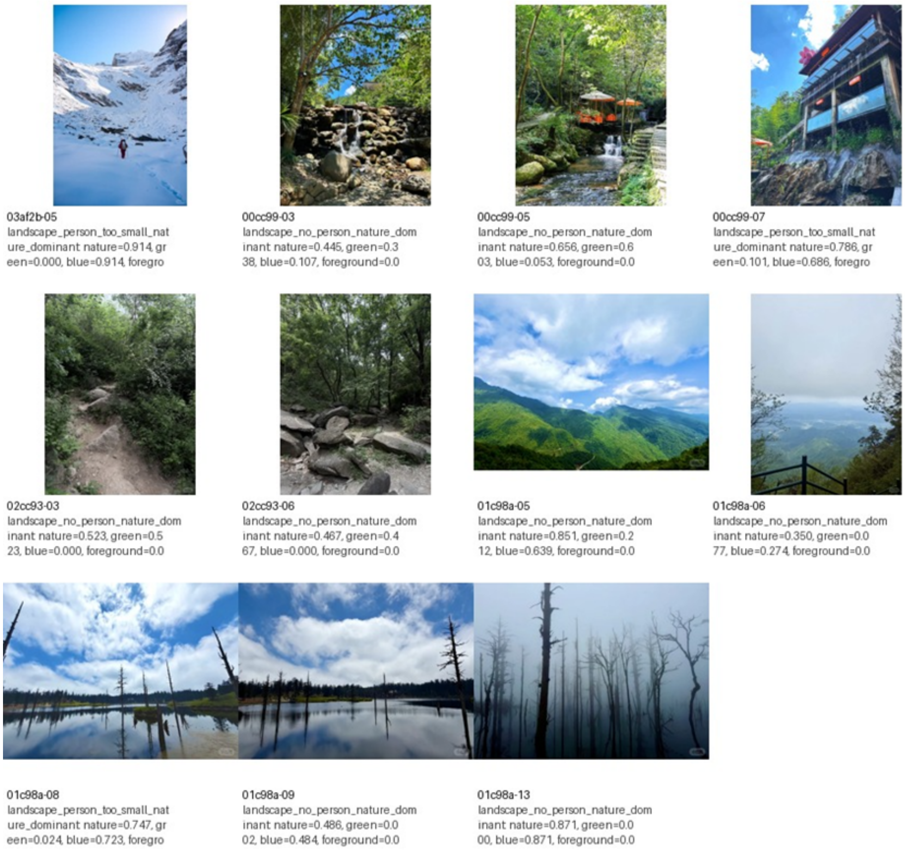

Figure 9. Partial landscape images filtered out by the gatekeeper.

6.2 Node B: Vision Large Model Annotation Module

The task of Node B is to perform structured visual annotation on outfit images that have passed Node A screening. This module uses the vision-language large model Qwen-VL-Max, called through the OpenAI-compatible interface of Alibaba Cloud Bailian DashScope [7]. The model has strong image understanding and visual semantic analysis capabilities. It can identify garment silhouettes, functional components, colours, material textures, and scene attributes, making it suitable for analysing Gorpcore functional-style outfits, which contain many visual design elements [6, 7].

The core principle of Node B is to constrain the output of the vision large model through prompt engineering. As illustrated in Figure 11, images in Xiaohongshu are diverse, including full-body outfits, half-body outfits, clothing-detail images, multi-image collages, image-text mixed content, and even product advertisements. The prompt-engineering design adopts a two-stage process—curation first, then annotation. In the first stage, the model must qualitatively determine whether

Figure 10. Qwen-VL-Max multimodel large model.

the image is a real outfit image or a pure landscape/text screenshot. Only when the first-stage curation status is judged as `use` is the model allowed to continue extracting specific attributes in the second stage. The second stage further standardises how different image types are handled (full-body, half-body, close-up, and collage). Clear rules are defined: only the clearest core outfit is analysed in a collage, product-only images without real model wearing are rejected, and local close-ups must not be used to infer the overall outfit. By converting complex edge cases into standardised processing logic, Node B improves the purity, quality, and reliability of the final structured labels.

(a) Half-body shots **(b)** Image-text mixed content **(c)** Multi-image collages

Figure 11. In addition to the standard full-body photos, the dataset also contains various types of pictures.

To allow Node C to directly read and process the model results, the system injects control parameters into the API request to enforce a JSON response format. A representative output record is shown below:

Listing 1: Representative Node B JSON output for a single image.


```
{
  "Image_ID": "GRP-XHS-686b88720000000120334f2-01",
  "Curation_Status": "use",
  "Image_Category": "full_body_outfit",
  "Reject_Reason": "none",
  "Body_Coverage": "full_body",
  "Text_Overlay_Level": "none",
  "Outfit_Count": 1,
  "Main_Subject_Visibility": "clear",
  "Pockets": 2,
  "Zipper_Type": "Sealed",
  "Fit": "Loose",
  "Reflective": true,
  "Primary_Color": "black",
  "Secondary_Color": "olive green",
  "Material_Clue": ["matte nylon", "ripstop"],
  "Scenario": "Urban_Commute",
  "Visual_Weight": "lightweight",
  "Gorpcore_Relevance": 0.87,
  "Confidence": 0.82
}
```


This module also includes an error-handling mechanism. If annotation fails for an image, the failure is recorded in `output/annotation_errors.csv`. Successfully annotated images are saved one by one in `output/json_labels/`. This per-image saving method supports breakpoint resumption, so even if an API error occurs midway, already completed annotation results will not be lost. The model output is treated as structured annotation candidates, not final ground truth. Low-confidence labels, ambiguous material descriptions and images with multiple outfits will be checked through Node C and human review before entering the curated dataset.

6.3 Node C: Semantic Cross-Checking Module (In Progress)

The task of Node C is to check whether the visual labels of an image are consistent with the original text description. On social platforms such as Xiaohongshu, "recommendation-style copywriting" is common. For example, the text may say "lightweight summer commuting," while the clothing shown in the image may appear heavy, hard-shell, and highly protective. Such image–

text inconsistency can affect trend judgement, so semantic cross-checking is required. Node C reads the original `title_text` and `raw_text` corresponding to each image and then matches them against the visual labels generated by Node B according to predefined rules.

For example, if the text contains terms such as “lightweight,” “thin,” “summer,” “commuting,” or “breathable,” but the visual label shows `Visual_Weight = heavyweight`, or the material clues include heavy materials such as `fleece` or `thick shell`, then this record will be marked as a low-confidence record. The role of Node C is to ensure that data curation does not rely solely on the image model, but instead combines textual semantics with visual judgement, thereby reducing the impact of platform copywriting bias.

6.4 Node D: Human Review Module (In Progress)

The task of Node D is to process the low-confidence data flagged by Node C and implement a Human-in-the-Loop mechanism. The course requirements emphasise that AI can assist data curation, but it cannot completely replace human judgement. Therefore, human review is an important component of this agent.

Node D displays the relevant information of low-confidence images one by one, including the `Image_ID`, `image_path`, original text, model labels, and reason for low confidence. Human reviewers can choose to `accept` the model labels, `edit` the labels, or `reject` the image entirely.

This file is the staged annotation result of Dataset v1 and contains image traceability information, visual labels, semantic-checking results, and human-review status. The design of Node D reflects the idea of human–AI collaborative data curation: AI is responsible for batch processing and preliminary judgement, while humans are responsible for the final confirmation of low-confidence samples.

7. Next Steps

The remaining work is organised into four tasks across Weeks 6–9, each with a clear owner-set, method and deliverable, as summarised in Table 3.

7.1 Feasibility and Risk

The plan is realistic because the hardest infrastructure—four collectors, a metadata schema, and working agent nodes—already exists and has produced real outputs. The main residual risks are vision-model cost at full scale, visual-clustering interpretability, and the standard schedule pressure of the final two weeks. Where a heavy step proves impractical, we have a lighter fallback—for example a rule-plus-ontology version of the visual word cloud if full clustering is too costly [5].

Table 3. Planned tasks and deliverables for Weeks 6–9.

Task	Main Objective	Key Work	Deliverable
T1. Full-corpus curation	Scale the agent from pilot to production	Run Node C cross-check; face-blur for v2	Curated Dataset v2 + automated curation report
T2. Visual word cloud	Build the image-based trend representation	Crop design components; embed with a visual model; cluster and render a data-driven mood board	Visual trend atlas / word cloud
T3. Dashboard & analysis	Make findings explorable	Streamlit dashboard for colour, material, pain-point and sentiment filtering; correlation analysis	Interactive curation dashboard
T4. Design strategy & final report	Turn evidence into recommendations	Synthesise pain-point→design mapping into a lightweight Gorpcore brief; write and rehearse final delivery	Design white paper + final report + slides

7.2 Engagement and Responsiveness

As shown in Figure 12, our team uses a shared GitHub repository to coordinate project writing, code development, data processing, and output management. The repository provides a central workspace where members can update their assigned sections, upload scripts, record intermediate results, and track changes throughout the project. This collaborative workflow helps ensure that the report, agent modules, datasets, and visual materials remain consistent, traceable, and easy to review. By working through GitHub, the team can manage version control more effectively and integrate individual contributions into a unified project outcome.

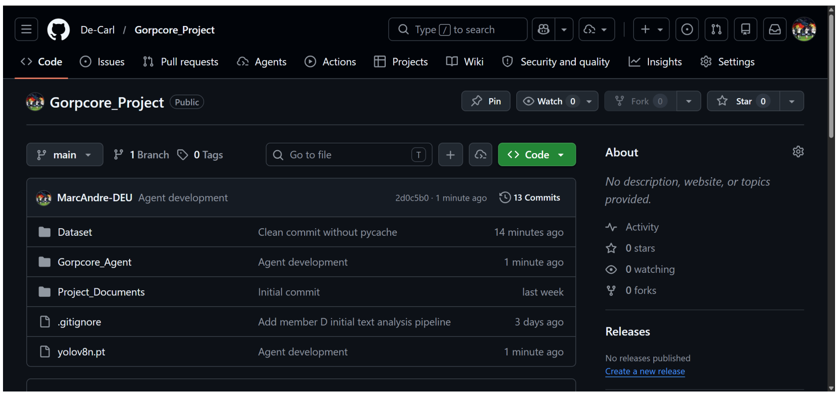

Figure 12. Github Repository for Team Collboration.

8. Conclusion

At the Week 6 checkpoint, the project has moved decisively from concept to a working, auditable curation system. We collect from four platforms, structure the data under a realised metadata schema with versioning, and run a multi-node agent that filters images, produces uniform visual annotations, analyses tens of thousands of text records, and translates consumer criticism into design implications—all while keeping a human in the loop. The early findings suggest a plausible direction for a lightweight, commute-ready Gorpcore line, especially around price perception, breathability, weight, fit and zipper reliability. However, these should remain provisional until full image annotation, semantic cross-checking and manual validation are completed. The remaining work is largely scaling and synthesis, which positions the team well for a strong final delivery in Week 9. In short, Gorpcore curation here is less about chasing what is popular and more about building a reusable, evidence-based bridge between messy public data and a defensible design decision.

References

[1] Group 12 Project Repository (code, data and outputs). https://github.com/De-Carl/Gorpcore_Project

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Appendix

Realised Micro-Ontology

The Text-Analysis Module normalises free text against a six-facet controlled vocabulary so that signals from different platforms become comparable. The facet sizes in the current version are listed in Table 4; the full term lists are in `micro_ontology.json`.

Table 4. Micro-ontology facets and representative terms.

Facet	Terms	Examples (translated)
Colour	19	olive / army green, khaki, charcoal, off-white, navy, contrast colour
Material	18	Gore-Tex, ripstop, nylon, fleece, soft-shell, hard-shell, Cordura, coating
Cut / silhouette	13	oversize, slim, cropped, longline, drop-shoulder, cinched waist
Functional component	16	taped seams, reflective, detachable, multi-pocket, sealed, lightweight, breathable, quick-dry
Scenario	15	commute, city, hiking, travel, ski, rain, camping, cycling, daily
Pain point	8	not breathable, too heavy, poor fit, zipper issue, high price, not durable, not commute-friendly